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A modified Sage-Husa adaptive Kalman filter for state estimation of electric vehicle servo control system

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Abstract

Kalman filter algorithm is widely used in state estimation of electric vehicles (EV) servo system as dynamic estimation of power system. In order to improve the estimation effect of noise interference in engineering practice, a scheme of Modified Sage-Husa adaptive Kalman filter (MSHAKF) combined with fuzzy clustering algorithm is proposed. The scheme processes the experimental data in a robust and adaptive way through the combination of constantly changing fuzzy clustering and MSHAKF, so as to better represent the dynamic characteristics of EV. The simulation results show that this method has higher estimation accuracy than the traditional method on the premise of unknown disturbance and noise characteristics in the system, and is more in line with the practical application of EV.

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Keywords: Sage-Husa adaptive Kalman filter; Fuzzy clustering algorithm; Servo system; Electric vehicle; Uncertain noise identification

1. Introduction

In the application of rotating instruments in industry, the research fields of mechanical structure optimization, control chattering suppression and fault diagnosis and classification all need to identify and control the dynamic change of exciting force on mechanical structure. Among them, the representative is the vibration control of modern EV. Permanent magnet synchronous motor (PMSM) has been one of the main power sources of EV. With the lightweight design of vehicles, the vibration problem of electric vehicle motors is becoming more and more prominent [1]. As an important part of EV, high performance PMSM system has high demand. For EV, special attention must be paid to vibration when high performance PMSM realizes basic operation functions [2]. It is not enough for the PMSM control system to only ensure high-precision operation, but also to maintain low vibration and noise during operation [3].

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It can be seen that noise suppression is the key to reduce the recognition error and improve the recognition accuracy in the servo motor control of EV. Kalman filter is used to suppress noise. Recursive least squares algorithm uses the gain, innovation and covariance matrix generated by Kalman filter to estimate the uncertain noise. Then, the numerical simulation of nonlinear system interference noise identification is carried out by combining extended Kalman filter (EKF) and recursive least square method [4,5]. Lin [6] designed an adaptive decreasing coefficient for the weight formulation of the least squares estimator to improve the accuracy and robustness of the estimator. Nazari and Pashazadeh [7] used fuzzy density clustering algorithm and Kalman filter to estimate the tracking of shipboard radar ship's movement position. The above research is carried out under the condition of determining the system parameters, that is, the mathematical model and the statistical characteristics of noise are known. However, it is difficult to obtain accurate mathematical model and noise statistical characteristics in practical application, which will reduce the filtering accuracy and even cause filter divergence.

Sage-Husa Adaptive Kalman Filter (SHAKF) [8–10] is an adaptive Kalman filter developed to deal with uncertain noise generated during measurement. SHAKF algorithm can estimate the statistical characteristics of noise in real time to reduce the filtering divergence. Its approach is to introduce the time noise observer into the Kalman filtering model. Based on the adaptive factor in the AKF model, SHAKF is further studied to improve the practicability and optimization of the filter. Aiming at the negative influence of abnormal values in the control signal on the controller, Lyu et al. [11] and Li et al. [12] proposed an adaptive robust AKF (ARKF), which can also make the system with fast switching of control signals converge quickly. The MSHAKF is a maximum likelihood estimation algorithm for unknown noise characteristics, which can estimate the characteristics of system process noise and online measurement noise [13–15]. The purpose of this paper is to identify the interference in the operation of EV when the statistical characteristics of noise and disturbance are unknown and the system model is not accurate enough.

In this paper, a combination of modified Sage-Husa adaptive Kalman filter and adaptive fuzzy clustering algorithm is proposed to identify the uncertain disturbance of the electric vehicle servo control system. In order to verify the performance of the proposed scheme, random noise and drift errors are introduced into the control signal of the electric vehicle servo system to verify the effect of noise estimation and error tracking on speed parameters. Compared with the commonly used EKF, the method proposed in this paper has obvious advantages in noise estimation performance.

2. The proposed method

2.1. Modified Sage-Husa Kalman filtering

By and large, the state estimation can be obtained by using the conventional Kalman filter algorithm for the mathematical model and statistical characteristics of known noise. However, in practical application, there are a lot of uncertain nonlinear factors, which makes it difficult to obtain accurate mathematical model and noise statistical characteristics, which will reduce the filtering accuracy, even lead to filtering shunt. Sage-Husa Kalman filter (SHKF) is a maximum likelihood estimation algorithm for unknown noise characteristics, which can estimate the characteristics of system process noise and measurement noise. This paper proposes an improved SHKF, which adds the function of on-line recognition and classification of noise.

State prediction can be obtained by predicting the presence of the state from the previous state:

$$\hat{\boldsymbol{X}}(k/k-1) = \boldsymbol{\Phi}\hat{\boldsymbol{X}}^*(k-1/k-1) + \hat{\boldsymbol{q}}(k) \tag{1}$$

The state of the system has been updated, and now the error estimation covariance matrix of the system needs to be updated:

$$\boldsymbol{P}(k/k-1) = \boldsymbol{\Phi}\boldsymbol{P}(k-1/k-1)\boldsymbol{\Phi}^T + \boldsymbol{\Gamma}\hat{\boldsymbol{X}}(k)\boldsymbol{\Gamma}^T \tag{2}$$

The difference between the best predicted value and the measured value (called innovation):

$$\varepsilon(k) = \boldsymbol{Z}(k) - \boldsymbol{H}\hat{\boldsymbol{X}}(k/k-1) - \hat{\boldsymbol{r}}(k) \tag{3}$$

It can be obtained innovation covariance:

$$\boldsymbol{S}(k) = \boldsymbol{H}\boldsymbol{P}(k/k-1)\boldsymbol{H}^T + \hat{\boldsymbol{R}}(k) \tag{4}$$

The Kalman gain is:

$$\boldsymbol{K}(k) = \boldsymbol{P}(k/k-1)\boldsymbol{H}^T\boldsymbol{S}^{-1}(k) \tag{5}$$

The state vector update law is:

$$\hat{\boldsymbol{X}}(k/k) = \hat{\boldsymbol{X}}(k/k-1) + \boldsymbol{K}(k)\varepsilon(k) \tag{6}$$

The update law of the covariance matrix is:

$$\boldsymbol{P}(k/k) = [\boldsymbol{I} - \boldsymbol{K}(k)\boldsymbol{H}]\boldsymbol{P}(k/k-1) \tag{7}$$

2.2. Adaptive fuzzy clustering technique

The uncertain function $\alpha_i(z_i, \theta_i)$ a can be expressed as a fuzzy system as follows:

$$\alpha_i(z_i, \theta_i) = \theta_i^T \varphi_i(z_i), i = 1, \ldots, m \tag{8}$$

where θ_i is the updated online parameters, which should be designed according to the actual application effect of the model. $\varphi_i(z_i)$ is the fuzzy basis function vector, as follows:

$$\varphi_i(x) = \frac{\left(\prod_{j=1}^{n} \mu_{A_j^i}(x_j)\right)}{\sum_{i=1}^{M}(\prod_{j=1}^{n} \mu_{A_j^i}(x_j))} \tag{9}$$

We can regard the input vector $x^T = [x_1, x_2, \ldots, x_n] \in R^n$ as the output vector $\hat{f} \in R$ obtained by the fuzzy inference engine using a set of IF–THEN, then the ith fuzzy rule can be expressed as

$$\text{IF} x_1 \text{ IS} A_1^i \text{ and } x_2 \text{ IS} A_2^i \text{ and} \cdots x_n \text{ IS} A_n^i \text{ THEN} f \text{ IS} f_i$$

using the above fuzzy rules, the output of the fuzzy system can be represented by the reasoning process and clustering center, as follows:

$$\hat{f}(x) = \frac{\sum_{i=1}^{M} f_i(\prod_{j=1}^{n} \mu_{A_j^i}(x_j))}{\sum_{i=1}^{M}(\prod_{j=1}^{n} \mu_{A_j^i}(x_j))} = \begin{bmatrix} f_1 & f_2 & \cdots & f_M \end{bmatrix} \varphi_i = \alpha_i(x_i, \theta_i) \tag{10}$$

From the above considerations, combined with the theory of SHKF, we can get

$$\begin{cases} \hat{\boldsymbol{X}}_k^i = \boldsymbol{A}^i \hat{\boldsymbol{X}}_{k-1}^i + \boldsymbol{B}^i \tilde{\boldsymbol{u}}_k^i + \boldsymbol{K}^i \varepsilon_{r_k}^i \\ \tilde{\boldsymbol{y}}_k^i = \boldsymbol{C}^i \hat{\boldsymbol{X}}_k^i + \boldsymbol{D}^i \tilde{\boldsymbol{u}}_k^i + \varepsilon_{r_k}^i \end{cases} \tag{11}$$

where $\boldsymbol{A}^i$, $\boldsymbol{B}^i$, $\boldsymbol{C}^i$, $\boldsymbol{D}^i$, and $\boldsymbol{K}^i$ represent state matrix, input state matrix, output correlation matrix, direct correlation matrix and Kalman gain matrix respectively, which are estimated by SHKF algorithm. The matrix $\tilde{\boldsymbol{Z}}_k = \begin{bmatrix} \tilde{\boldsymbol{u}}_k & \tilde{\boldsymbol{y}}_k \end{bmatrix}^T$ belongs to fuzzy set A_1^i with a value $\mu_{A_j^i}$ defined by a membership function $\mu_{A_j^i}: \mathbb{R} \to [0, 1]$, with $\mu_{A_j^i} \in \mu_{A_1^i}, \mu_{A_2^i}, \mu_{A_3^i}, \ldots, \mu_{A_j^i}$. The error of fuzzy reconstruction can be written as residual sequence, which is given by the following equation:

$$\varepsilon_{r_k}^i = \tilde{\boldsymbol{y}}_k^i - \gamma^i \boldsymbol{y}_k^i \tag{12}$$

where γ^i is a weight parameter that depends on the value of the ith rule member; $\boldsymbol{y}_k$ represents the real output at sample k; $\tilde{\boldsymbol{y}}_k^i$ is equivalent to $\alpha_i(x_i, \theta_i)$ in (10) and represents the output of SHKF in the ith rule at sample k.

The fuzzy clustering algorithm proposed in this paper is modified on the basis of the classical Gustafson–Kessel, and its initial value is determined by the clustering set established based on the previously collected data. The cluster prototype is shown in the following equation.

$$v_i^{(l)} = \frac{\sum_{k=1}^{N}\left[\left(\mu_{ik}^{(l-1)}\right)^m \boldsymbol{z}_k\right]}{\sum_{k=1}^{N}\left(\mu_{ik}^{(l-1)}\right)^m}, 1 \le \text{i} \le \text{c} \tag{13}$$

where z_k means the data in the used sample k and μ_{ik} is the membership degree of the ith cluster in the same sample. We can get the cluster covariance matrix F_i as follows:

$$F_i = \frac{\sum_{\mathrm{k}=1}^{N} \left[\left(\mu_{ik}^{l-1}\right)^m \left(\boldsymbol{z}_k - \boldsymbol{v}_i^{(l)}\right) \left(\boldsymbol{z}_k - \boldsymbol{v}_i^{(l)}\right)^T \right]}{\sum_{k=1}^{N} \left(\mu_{ik}^{(l-1)}\right)^m} \tag{14}$$

Gustafson–Kessel algorithm makes each cluster have its own norm induction matrix. The way is to judge the number of clusters with different geometries in the data set through the adaptive Mahalanobis distance norm detection mechanism. [16]. The inner product norm generated by the matrix is as follows:

$$D_{\mathrm{ik}O_{\mathrm{i}}} = \sqrt{(z_k - v_i^{(l)})^T o_i \left(z_k - v_i^{(l)}\right)} \tag{15}$$

So, we can update the partition matrix and get the following equation:

$$\mu_i^{(k)} = \frac{1}{\sum_{j=1}^{c} \left(\frac{D_{ikO_i}}{D_{jkO_i}}\right)^{2/(m-1)}} \tag{16}$$

Gustafson–Kessel clustering algorithm evaluates the similarity between new data points and each cluster by checking Mahalanobis distance. Mahalanobis distance is defined as follows:

$$d_{ik} = \sqrt{(\boldsymbol{z}_k - \boldsymbol{v}_i) \left[(\rho_i det(\mathbf{F}_i))^{1/n} \mathbf{F}_i^{-1}\right] (\boldsymbol{z}_k - \boldsymbol{v}_i)^T} \tag{17}$$

From the Mahalanobis distance, we can get the closest cluster as follows:

$$p = \arg\min (d_{ik}), \quad i = 1, \ldots, c. \tag{18}$$

The radius of Eq. (17) can be computed as follows:

$$r_p = \max \left\| \boldsymbol{v}_p - z_k \right\|_{o_p}, \forall z_k \in p\text{-}th \tag{19}$$

For the new data after clustering, we will evaluate it according to the following conditions

If $d_{ik} \leq r_p$, the newly generated data still belongs to the existing cluster. Then the center of the cluster and the correlation matrix will be updated. The update rules are as follows:

$$\begin{cases} \boldsymbol{v}_{p.new} = \boldsymbol{v}_{p.old} + \sigma\left(z_k - \boldsymbol{v}_{p.old}\right) \\ \mathbf{F}_{p.new} = \mathbf{F}_{p.old} + \sigma\left(\left(\zeta^T \zeta\right) - \mathbf{F}_{p.old}\right) \\ \zeta = \left(\boldsymbol{z}_k - \boldsymbol{v}_{p.old}\right) \end{cases} \tag{20}$$

where σ represents the learning rate of the current sample data; $\boldsymbol{v}_{p.new}$ and $\boldsymbol{v}_{p.old}$ represents the new value and the old value of the cluster center, respectively; In the same way $\mathbf{F}_{p.new}$ and $\mathbf{F}_{p.old}$ represents the values of the covariance matrix before and after updating.

If $d_{pk} > r_p$, this means that the updated data points do not belong to the existing cluster. We need to regard them as new clusters and redefine them.

$$\begin{cases} \boldsymbol{v}_{new} = \boldsymbol{z}_k \\ \mathbf{F}_{new} = \mathbf{F}_{p.old} \end{cases} \tag{21}$$

When defining a new cluster, the reliability of the cluster cannot be guaranteed. Therefore, we need to use a parameter to evaluate the number of points belonging to the cluster. This parameter is used as a threshold to represent the minimum number of data points required for the new cluster. If the number of data points in the new cluster is greater than this threshold, the new cluster is retained. Otherwise, refuse to establish a new cluster and maintain the original data structure.

In addition, we established a partition coefficient used to determine the optimal number of clusters as a way to evaluate the quality of fuzzy clustering, as follows:

$$PC\left(c\right) = \frac{1}{N}\sum_{i=1}^{c}\sum_{\beta=1}^{N}\left(\mu_{i\beta}\right)^{2} \tag{22}$$

where i represents the number of clusters and β is data point used to express the identity of a member in cluster i.

2.3. Parametric estimation

The proposed MSHAKF algorithm proposed in this paper is a direct Kalman filter gain method. It is similar to the AKF, it does not need prior statistical information, nor does it depend on the calculation of sample correlation or covariance. The steps of the adaptive fuzzy SHKF algorithm proposed in this paper are shown in Fig. 1.

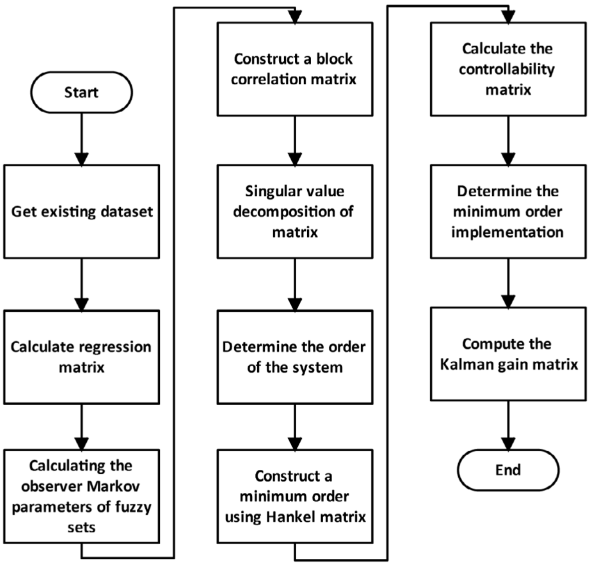

Fig. 1. Estimation process of MSHAKF algorithm.

2.4. Simulation

The purpose of this part is to predict and track the state of a group of data through simulation experiments to show the feasibility and effectiveness of the scheme proposed in this paper. The simulation is run using MATLAB 2020a on a personal computer with Intel Core, CPU i5-6500 @ 3.20 GHz, 16 GB RAM.

It can be seen from Fig. 2(a) that MSHAKF proposed in this paper has better tracking effect on data, especially where the data changes obviously, the adaptive parameter tuning of the scheme proposed in this paper has a more obvious effect. Compared with the conventional SHKF and the traditional EKF, the proposed method has significant advantages in prediction and tracking accuracy.

In addition, in order to prove that the advantages of MSHAKF are not caused solely by the increase of adaptive law, we supplemented the performance comparison simulation test with adaptive robust unscented Kalman filter (ARKF) [12]. As shown in Fig. 2(b), ARKF has strong performance in tracking control signals with uncertainty, However, Sage-Husa Kalman filter has stronger performance when dealing with vibration signals and small noise in electric vehicle servo system, which can be seen more clearly through root mean square error (RMSE).

3. Experimental results

This paper takes a PMSM servo controller applied to an EV as a case to verify the application effect of the method proposed in this paper. The PMSM mathematical model has three main parts: power grid, electromechanical torque and mechanical system.

The focus of this paper is to track the uncertain interference of EV in driving, so the following simplification is made in the mathematical model: ignoring saturation, the induced electromagnetic force is evenly distributed,

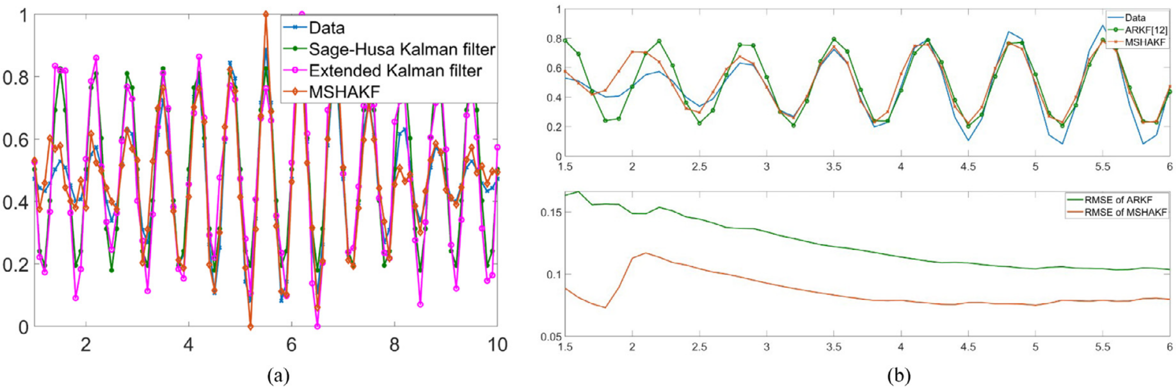

Fig. 2. Simulation results of data state estimation based on MSHAKF. (a) Comparison of three state estimation algorithms. (b) Comparison of two types of adaptive Kalman filter algorithms.

ignoring eddy current and hysteresis loss, no dynamic coupling in the air gap, no rotor cage. The rotor electrical equation of PMSM is as follows:

$$u_d = R_s i_d + L_d \frac{di_d}{dt} - p\omega_r L_q i_q \tag{23}$$

$$u_q = R_s i_q + L_q \frac{di_q}{dt} - p\omega_r L_d i_d + p\omega_r \Psi \tag{24}$$

where u_d and u_q represent the dq axis voltages, L_d and L_q represent the dq axis inductances, R_s and Ψ represent the stator resistance and the magnetic flux generated by the permanent magnet on the rotor, respectively.

Combined with drive dynamics, the equation of motion of PMSM can be written as:

$$\frac{dw}{dt} = \frac{p}{J}\left[\frac{3}{2}(W_m - (L_q - L_d)id) \cdot i_q\right] - \frac{T_l}{J} \tag{25}$$

where p represents the number of pole pairs, T_l and J is represent load torque and total moment of inertia, respectively.

According to the characteristics of the disturbance noise in the electric vehicle servo controller, we select the following equation to estimate the statistical characteristics of the noise.

$$\hat{\boldsymbol{r}}(k) = (1 - d_{k-1})\hat{\boldsymbol{r}}(k-1) + d_{k-1}[\boldsymbol{Z}(k) - \boldsymbol{H}\hat{\boldsymbol{X}}(k/k-1)] \tag{26}$$

$$\hat{\boldsymbol{R}}(k) = (1 - d_{k-1})\hat{\boldsymbol{R}}(k-1) + d_{k-1}[\varepsilon(k)\varepsilon^T(k) - \boldsymbol{H}\boldsymbol{P}(k/k-1)\boldsymbol{H}^T] \tag{27}$$

In order to test the behavior of observers with different working conditions, it was decided to use two speeds for testing, $\omega = 5$ rad/s and $\omega = 100$ rad/s, respectively.

The simulation architecture designed in this paper is mainly divided into three parts: driving behavior, vector control of PMSM and MSHAKF algorithm proposed in this paper. Their specific structure is shown in Fig. 3. We use different colors to mark and distinguish the above three parts. It should be noted that in the MSHAKF algorithm proposed in this paper, the adaptive gain cannot be solved in one iteration, and the optimal gain matrix needs to be determined by formulating an appropriate cost function.

The ideal control signal in Fig. 4(a) refers to the standard signal output by the controller, which includes two processes of starting and braking. The real signal acquisition is the data collected by the EV sensor when the electric vehicle is running in the corresponding situation, and there are a variety of uncertain interference noises. By comparing the EKF [17] with the proposed method, it is obvious that the state estimation effect of the proposed method is better than that of the classical extended Kalman filter.

Fig. 4(b) shows the effect of state estimation at another speed. It can be seen that the response of EKF is more intense in the low-speed range or high-speed range under the environment of uncertain interference. The method proposed in this paper can estimate the state of EV with better tracking error.

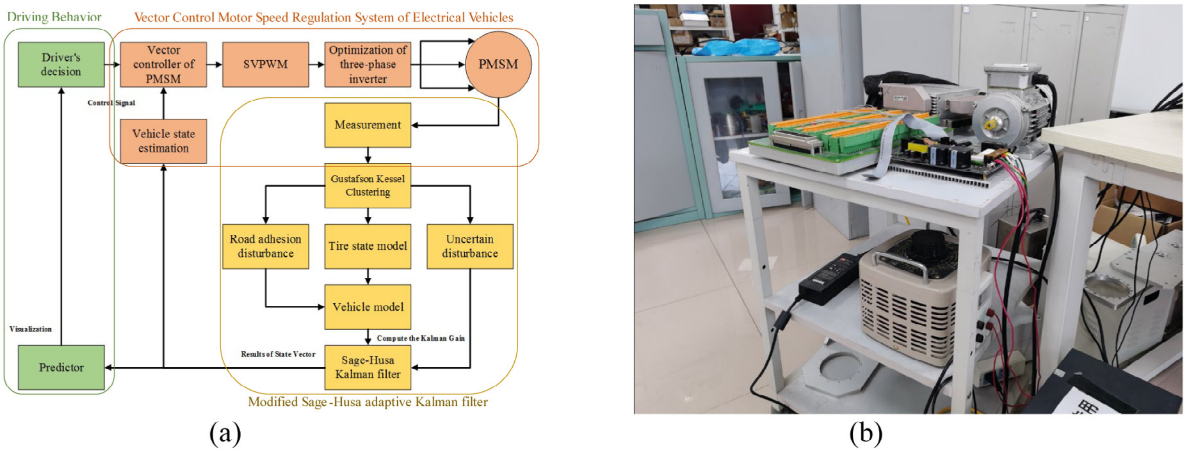

Fig. 3. Experimental platform for state estimation of electric vehicle servo control based on MSHAKF. (a) Block diagram of MSHAKF in servo control. (b) Hardware in the loop simulation experiment platform. . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

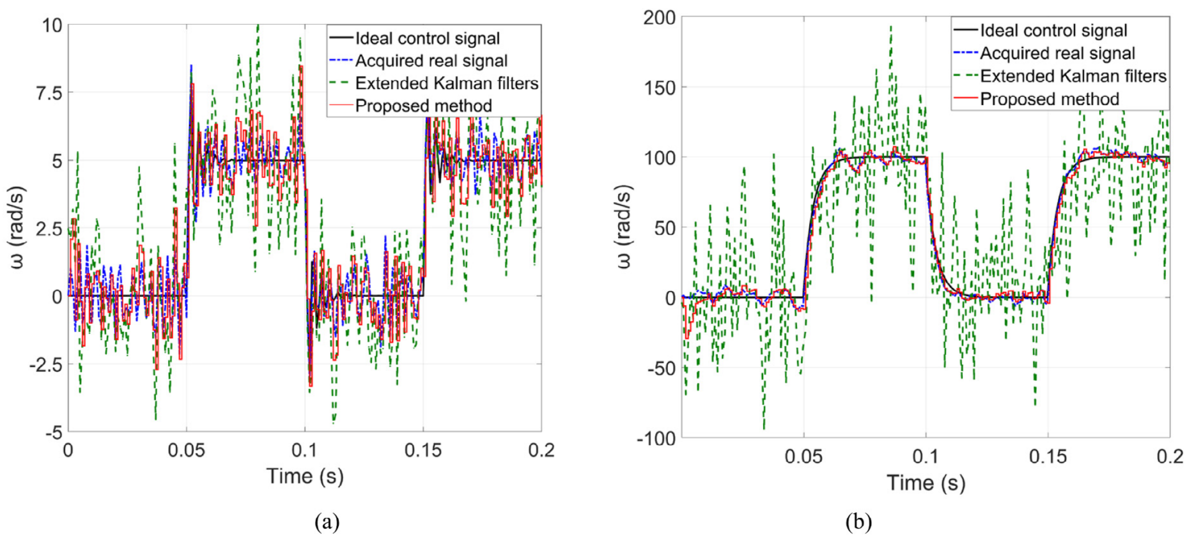

Fig. 4. The result of comparing the estimated error between EKF and the proposed method. (a) In the case of $\omega = 5$ rad/s. (b) In the case of $\omega = 100$ rad/s.

4. Conclusion

In this paper, a modified Sage-Husa adaptive Kalman filter is proposed for the estimation of disturbance and noise in dynamic nonlinear systems. In the MSHAKF algorithm proposed in this paper, the fuzzy adaptive clustering algorithm is applied to update the cluster center in each iteration, and the parameters are adaptively adjusted when the new cluster center appears. It is used to estimate the estimated state and output of the dynamic system, and changes the Kalman filter structure through the adaptive mechanism to make it possible to correspond to the state of the time-varying dynamic model in time. In the case given in this paper, it can be seen that MSHARKF is suitable for estimating the noise and chattering produced by the electric vehicle servo system in the process of operation and measurement. We have observed the noise estimation and error tracking effect of the MSHAKF algorithm on the speed signal at two speeds. Compared with the classical EKF algorithm, the proposed scheme can estimate the actual signal of the controller more accurately. During the experiment, we found that the algorithm proposed in this paper cannot deal with high-frequency pulse interference. How to deal with multiple types of mixed noise in the controller system will be the direction we need to continue to explore in this field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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